Unmesh Bhosale

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Education

University of Massachusetts Amherst — Amherst, MA
Master of Science in Computer Science — Jan 2026 – Dec 2027 (Expected)
Coursework: Machine Learning, Algorithms for Data Science, Distributed and Operating Systems

Sinhgad College of Engineering — Pune, India
Bachelor of Engineering in Electronics and Telecommunication Engineering — Aug 2019 – Jul 2023
Coursework: Data Structures and Algorithms, Deep Learning, DBMS, Operating Systems, Microcontrollers

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++ familiarity
Machine Learning / Deep Learning: PyTorch, NumPy, pandas, scikit-learn, JAX basics, neural networks, model evaluation, feature engineering, cross-validation, optimization algorithms
GenAI / Search / Vector Systems: natural-language search, intent parsing, content-based filtering, recommendation workflows, retrieval pipelines, vector search fundamentals, semantic search fundamentals
Backend / Big Data Systems: REST APIs, Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, distributed systems, socket programming, multithreading, caching, fault tolerance, Kafka/Spark fundamentals
Tools: Git, Linux, Docker, matplotlib, benchmarking, debugging, performance optimization

Projects

JAX-Based Neural Network Optimization and Semantic Retrieval Study | *Python, JAX, NumPy, FAISS* — *May 2026 – Present*

- Extended a NumPy-based neural network into JAX using `jax.numpy`, `jax.grad`, and `jax.jit` to study automatic differentiation, vectorized training, JIT compilation, and ML workload optimization.
- Integrated a FAISS-based vector search component to experiment with embedding similarity, nearest-neighbor retrieval, and semantic search workflows over structured ML experiment results.
- Benchmarked NumPy, eager JAX, and JIT-compiled JAX training loops using runtime measurements, convergence curves, and throughput-style evaluation.
- Strengthened practical foundations in scalable AI model training, optimization, vector search, and deployment-aware ML performance analysis relevant to GenAI and large-scale AI systems.

Neural Network Training Pipeline from Scratch | *Python, NumPy, pandas, matplotlib* — *Apr 2026 – May 2026*

- Built a modular neural-network training pipeline from scratch using NumPy, including forward propagation, backpropagation, binary cross-entropy loss, gradient updates, and L2 regularization.
- Implemented preprocessing, stratified 5-fold cross-validation, model training, and evaluation workflows across multiple datasets without using machine-learning model libraries.
- Evaluated architectures using accuracy, precision, recall, F1-score, learning curves, and convergence plots to debug training behavior and improve model selection.

Distributed Marketplace System Simulation | *Python, Sockets, Multithreading* — *Mar 2026 – Apr 2026*

- Built a distributed marketplace simulation with buyer, seller, trader, and warehouse services using Python sockets and multithreading.
- Implemented caching, leader election, fault tolerance, and timestamped logging to improve reliability across distributed components.
- Created benchmark scripts to compare cache-enabled and cache-less execution, analyzing performance, scalability, and failure handling for backend systems similar to large-scale AI infrastructure.

Professional Experience

AutoAgent – Co-Founder — *Pune, India | Apr 2024 – Jan 2026*

- Co-founded an IT solutions startup and delivered AI-driven web and backend solutions to 37+ global clients across e-commerce and small business domains.
- Designed an AI-powered e-commerce assistant using React, Node.js, Express.js, and MongoDB, enabling natural-language product search, intent parsing, dynamic filtering, and recommendation workflows.
- Optimized product retrieval and query workflows, reducing product search time by 60% while improving search relevance, user engagement, and conversion-oriented product discovery.
- Collaborated directly with clients, developers, and product stakeholders to translate business requirements into reliable AI backed web applications.

Search-In – Full Stack Web Developer Intern — *Pune, India | Oct 2023 – Mar 2024*

- Built full-stack product features including authentication, dashboards, REST APIs, and payment integrations using React, Node.js, Express.js, and MongoDB.
- Improved backend reliability and API responsiveness by debugging integration issues, optimizing database queries, and refining request-handling workflows.
- Collaborated in a fast-paced product environment to ship user-facing features with focus on usability, reliability, and clean frontend/backend integration.

Certifications

Mathematics for Machine Learning and Data Science — *May 2026 – Present*
Web Development: React and Backend with Node.js — *Jan 2025*